

Lecture 4: an introduction to vectors.

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Last time: We learned there are at least 2

sizes of infinity:

Countable: "# of whole numbers" = $|\mathbb{N}| = |\mathbb{Z}| = |\mathbb{Q}|$

Uncountable: "# of numbers between numbers, aka the reals" = $|\mathbb{R}|$

Most importantly: $|\mathbb{N}| = |\mathbb{Z}| = |\mathbb{Q}| < |\mathbb{R}|$

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Vectors: type of number system with magnitude

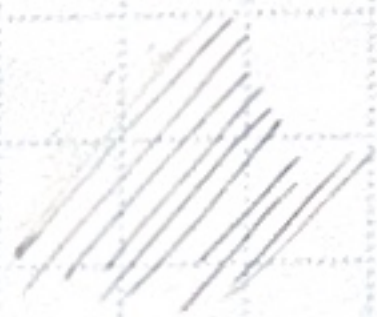
and direction. We often write vectors with arrows above letters, for example a vector V is written as $\vec{V}$

What is a magnitude?

Magnitude is a generalization of size or length.

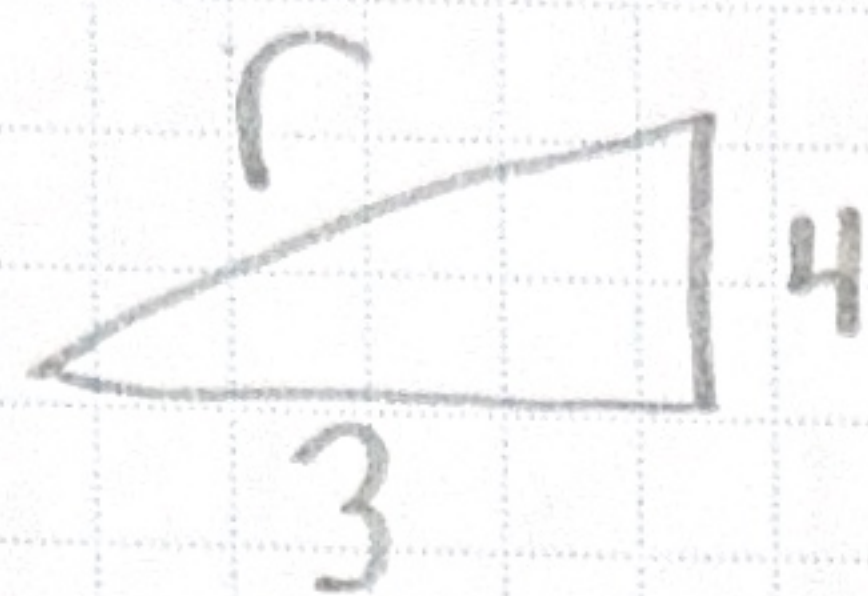
Consider how S and $-S$ both have absolute values of S , $|S| = |-S| = S$, one can say they both have magnitude of S .

Magnitude $\hat{=}$ size



Or consider:

the length of r on this triangle:



$$r^2 = 3^2 + 4^2 = 25 = 5^2 = r^2 \Rightarrow r = 5$$

r has length of 5

magnitude $\hat{=}$ length $\hat{=}$ size

Magnitude captures the general idea of size and length into one concept. We denote

the magnitude of a vector $\vec{v}$ with double double bars.

$$\|\vec{v}\| = \text{magnitude of } \vec{v}.$$

direction:

intuitively, this is where the vector points

See: dimension

We often use left/Right (or $+/-$) for 1 dimensional vectors (also called scalars)

up/down, L/R for 2D

up/down, L/R, in/out for 3D
(fore/back)

4d and up is hard to visualize.

But what is a dimension?

Dimension can be thought of as the # of unique Basis vectors needed to describe a vector.

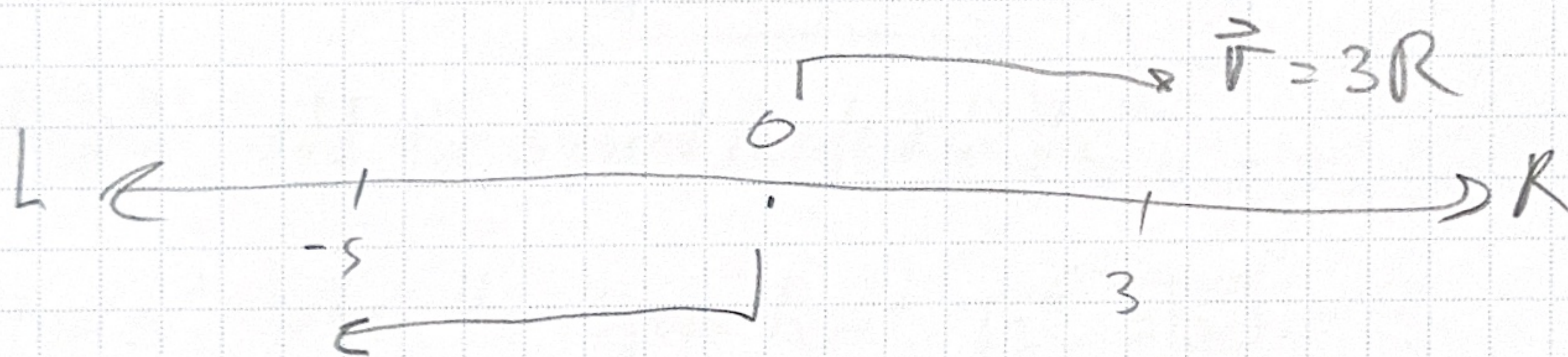
↳ the idea of 1D is a Basis, so to describe a point on a line you only need to describe how far L/R to go



go Left

$$\vec{v} = 1L, \text{ where } L \text{ is a Basis}$$

or: $\vec{v} = -1R$ since right is the opposite of Left.



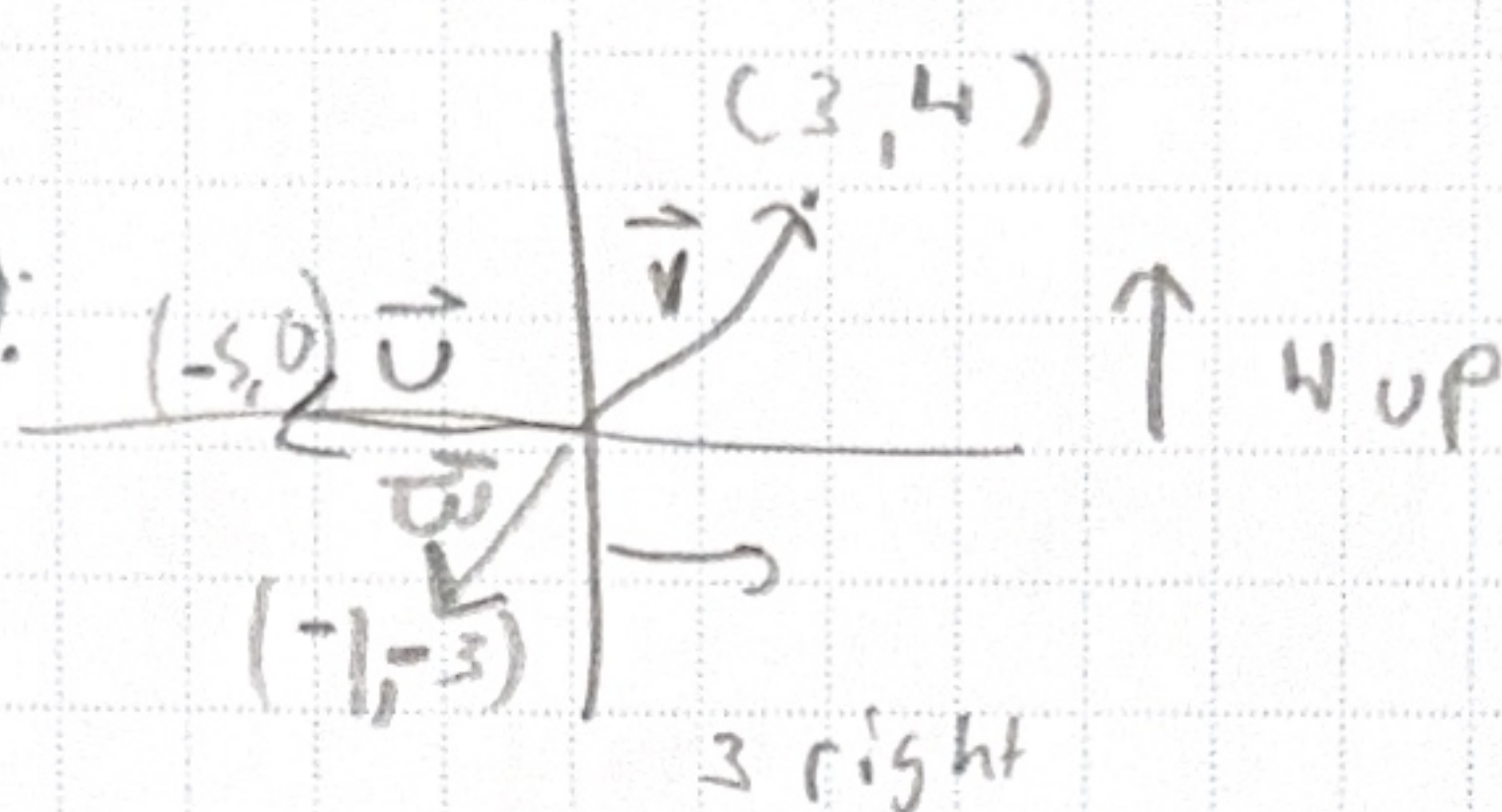
$$\vec{v} = 5L = 5(-R) = -5R$$

So we can reach all places on the line with one Basis,

namely R, thus this is a 1 dimensional vector space,
aka, a system where vectors operate in.

in 2 dimensions:

add in a up component U:



$$\vec{v} = 3R + 4U$$

$$\vec{w} = -1R + (-3)U = -1R - 3U$$

$$\vec{u} = -5R + 0U$$

Finding magnitudes of vectors:

Step 1: take the square of the component vectors
(you can ignore signs)

$$\vec{v} = (3)R - (4)U$$

$\swarrow$ $\swarrow$
 3^2 4^2

Step 2: add the squared numbers: $3^2 + 4^2 = 25$

Step 3: take the square root $\sqrt{25} = 5$

Note: this works for 1D and N dimensions.

$$\sqrt{a^2} = |a|$$

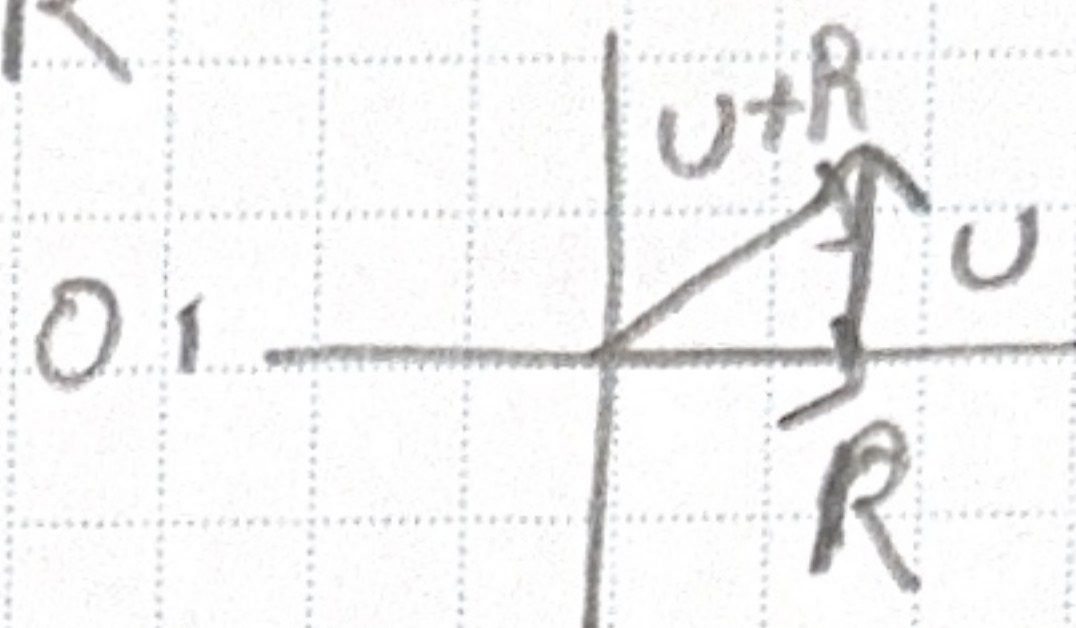
$$\sqrt{\sum a_i^2}$$

Properties of vectors:

additivity: $\vec{V} + \vec{U}$ can be expressed as a 'tip to tail' sum of vectors.

if $\vec{V} = U$ and $\vec{U} = R$

$$\vec{V} + \vec{U} = U + R$$



Rules for adding vectors:

For each component of $\vec{V}$ and $\vec{U}$, add them together to get $\vec{V} + \vec{U}$. For example! if

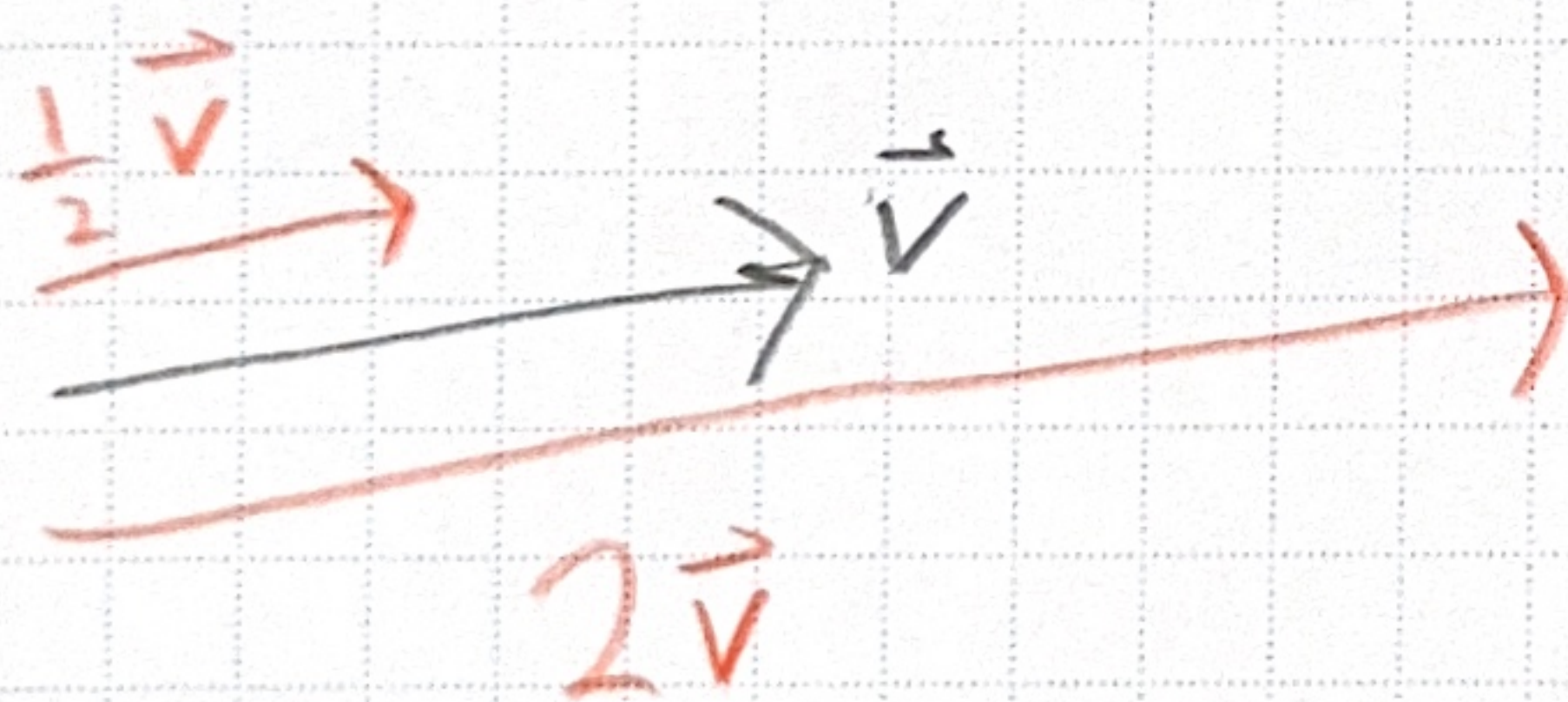
$$\vec{V} = 5R - 3U$$

$$\vec{U} = 2R + 1U$$

$$\begin{aligned}\vec{V} + \vec{U} &= (5+2)R + (-3+1)U \\ &= 7R - 2U\end{aligned}$$

Scaling By a scalar

$\lambda \vec{V} = \vec{V}$ stretched or shrunk by λ (lambda)



These two properties are called

Linearity

Which is a top 5 property of all time.

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